
Too Much to Handle: Retrieval-Augmented Prompts Degrade Factuality in Small Language Models—and Defeat Lookback Lens Hallucination Control

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Abstract

While retrieval-augmented generation (RAG) and attention-based decoding have emerged as promising approaches to reduce factual hallucinations in language models, little is known about their interplay—especially in small-scale models deployed in constrained settings. This study investigates whether attention-based hallucination detection, such as Lookback Lens, remains effective when external knowledge is introduced through retrieval. We implement **RELF (Retrieval-Enhanced Lookback Fusion) Decoding**, a decoding strategy that augments prompts with Wikipedia passages and reranks candidate completions using internal attention-based signals. Applied to a 7B LLaMA-2 model on the Natural Questions Open benchmark, RELF was evaluated alongside standard greedy decoding, standard Lookback Lens decoding, and their respective RAG-augmented counterparts. Contrary to expectations, adding retrieved context degraded factual accuracy across the board. Exact match performance dropped from 71.8% (Lookback) to 38.5% (RELF). Analysis reveals two compounding factors: the small model’s limited ability to process long, injected passages, and the retrieval component’s tendency to return semantically relevant yet answer-irrelevant documents—both of which undermine attention-based estimation. Our findings show that simply combining retrieval with attention-based detectors does not ensure better factuality in small models. Instead, effective hallucination mitigation in this setting requires retrieval-aware confidence mechanisms and capacity-sensitive context integration.

1 Introduction

Small Language Models (SLMs) are increasingly favored for deployment in resource-constrained and privacy-sensitive environments such as mobile devices, hospital intranets, and edge computing platforms [1]. These compact models enable local inference tasks in resource-constrained environments—such as process automation systems, medical summarization pipelines, or privacy-sensitive deployments—eliminating the need to transmit sensitive data to external servers.

However, their limited parameter capacity often leads to factual hallucinations, where the model generates confident yet incorrect information due to an incomplete internal representation of world knowledge [2].

To address hallucinations, previous work has explored retrieval augmented generation (RAG) [3], which supplements input prompts with relevant passages from large knowledge sources like Wikipedia. By providing external evidence, RAG aims to anchor generation to factual content and reduce hallucination. Meanwhile, methods like Lookback Lens [4] leverage internal attention signals to assess token-level reliability during generation. While lightweight and model-agnostic, these approaches lack external grounding and can still produce plausible but false statements when context is weak or misleading.

In this work, we hypothesized that combining external retrieval with attention-based hallucination scoring could yield complementary benefits. We implemented a RELF decoding strategy that fuses Wikipedia-retrieved context with Lookback-guided candidate selection. RELF comprises two complementary modules:

Lookback Scores (internal attention coherence): During decoding, we generate multiple candidate responses. For each candidate, token-level attention patterns from all transformer layer and head are used to compute Lookback features, which are fed into a classifier predicting the likelihood of hallucination-free output.

Retrieval-Augmented Context (external evidence grounding): Before decoding, we retrieve relevant Wikipedia passages based on the query using dense sentence embeddings, and prepend them to the prompt. This is intended to provide external factual grounding.

Unexpectedly, incorporating RAG lowered factual accuracy in both the Greedy + RAG and Lookback + RAG (RELF) configurations, increasing—rather than suppressing—hallucinations. We attribute this decline to the 7-billion-parameter model’s limited capacity to reason over the long, retrieval-augmented context.

Through qualitative case studies and corpus-level metrics, we observe that most failures trace back to the SLM’s inability to *leverage* the retrieved context. Two factors dominate: (i) the augmented prompt often exceeds the model’s effective context window, and (ii) the retriever frequently returns passages that are semantically similar to the query yet omit the correct answer altogether. Consequently, hallucinations in SLMs stem not only from missing internal knowledge, but also from an impaired capacity to filter and focus on the few tokens that truly matter amid noisy or misleading evidence.

These findings expose a critical caveat for RAG in small-model regimes: *more context is not necessarily better*. Factuality depends on how the model allocates attention across the original prompt, the retrieved passages, and its own emerging generation. When that allocation falters—because the context is too long or because retrieval quality is poor—hallucinations proliferate.

2 Related Work

Hallucination in Language Models Factual hallucination—where models generate fluent yet untrue statements—is a persistent issue in language generation tasks such as summarization and open-domain question answering. While large-scale language models mitigate this issue to some extent through greater parameter capacity and implicit knowledge, SLMs are more vulnerable due to their limited ability to encode and recall factual content. Prior work has approached hallucination detection through prompt tuning, confidence scoring, and filtering strategies, but these often require substantial model capacity or multi-step post-processing[5, 6, 7, 8].

Retrieval-Augmented Generation Retrieval-augmented generation (RAG)[3] incorporates external documents during generation to improve factual grounding[9, 10]. By conditioning generation on retrieved passages—typically from knowledge sources like Wikipedia—these systems enhance the factual correctness of responses. However, conventional RAG pipelines rely on large dense retrievers and may involve multiple stages of document ranking and verification, which impose non-trivial computational overhead. This makes them less suitable for deployment in low-resource or on-device settings.

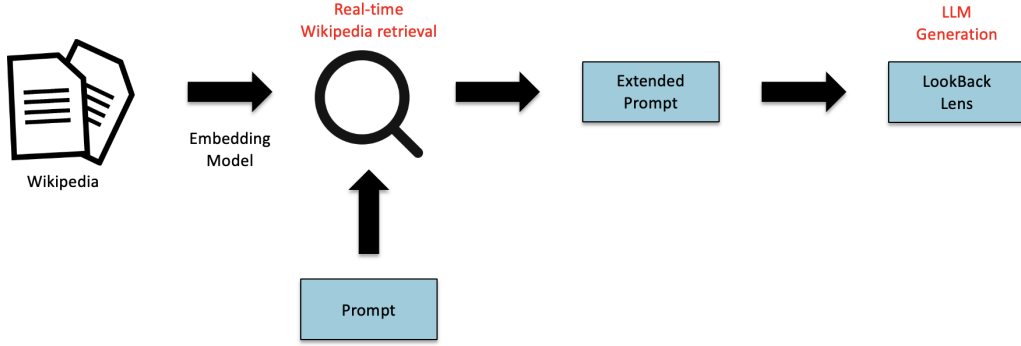


Figure 1: Overview of RELF: Real-time Wikipedia retrieval augments the prompt prior to generation. The extended prompt is decoded by the SLM and scored by Lookback Lens for hallucination likelihood.

Attention-Based Hallucination Detection Recent studies have shown that attention patterns within a language model can be indicative of factual consistency. In particular, attention-based decoding methods have demonstrated the ability to estimate hallucination likelihood by measuring how well a token is supported by the model’s prior context[4]. These techniques are lightweight and require no additional model calls, making them attractive for constrained environments. However, they remain limited when the input context itself lacks sufficient factual content, resulting in overconfident but unsupported generations.

Hidden State-Based Hallucination Detection. In parallel to attention-based methods, several approaches have explored the use of hidden state dynamics to detect hallucinations[11, 12, 13]. These techniques analyze internal representations (e.g., layer-wise hidden states) to estimate the factuality of a generated token. For example, Duan et al. [12] proposed to monitor changes in hidden state trajectories during generation, identifying patterns that correlate with hallucinated content. Such methods can be more expressive than attention-only approaches, capturing deeper semantic inconsistencies. However, they often require access to full model internals and may be less interpretable or efficient in practical decoding settings.

3 Approach

We implement **RELF (Retrieval-Enhanced Lookback Fusion) Decoding**, a decoding framework initially designed to reduce factual hallucination in small-scale language models. RELF integrates two complementary components: internal attention-based confidence signals, as used in the Lookback Lens framework, and external evidence retrieved in real time from Wikipedia. While RELF was intended to steer generation toward factually accurate outputs, our findings suggest that this fusion can be counterproductive in input-context-based tasks. This section describes the design of RELF and highlights its limitations revealed through experimentation.

3.1 Problem Setup

Given an input query q , such as a natural language question, our goal is to generate a response $\hat{y}$ that is not only fluent and contextually coherent but also factually grounded in relevant knowledge. For open-domain QA, where the input is often a single question without extended context, we hypothesized that supplementing the prompt with retrieved passages would reduce hallucination by anchoring generation to external evidence.

3.2 RELF Decoding Framework

RELF operates in three stages:

1. **Retrieval-Augmented Prompting:** We retrieve k Wikipedia passages $R = \{r_1, r_2, \dots, r_k\}$ using a dense retriever (E5-base) based on the input query q . The prompt is then constructed as:

$$\text{Prompt} = \text{concat}(r_1, r_2, \dots, r_k, \text{"\#Question\#:"}, q, \text{"\#Answer\#:"})$$

This step was intended to provide external factual grounding, especially in cases where internal model knowledge is insufficient.

2. **Candidate Generation:** The language model then generates n candidate outputs $\hat{y}_1, \dots, \hat{y}_n$ using top- p sampling with a temperature-controlled decoding strategy.
3. **Attention-Based Scoring and Selection:** In the original Lookback Lens Guided Decoding [4], candidate continuations are scored using averaged lookback ratios computed from attention heads across layers, and a logistic regression classifier is used to predict whether a token span is likely to be hallucinated. This classifier operates over features derived purely from attention maps, not token content, enabling lightweight hallucination detection based on how much attention is paid to previous context. For a detailed explanation of the Lookback Lens mechanism, including the definition of lookback ratios and the logistic regression classifier, please refer to Appendix B.

Specifically, we compute the lookback attention features for each candidate $\hat{y}_i$ by aggregating lookback ratios across all transformer layers and heads, following the original Lookback Lens formulation. These features are fed into a pretrained logistic regression classifier f_θ :

$$s_i = f_\theta(\text{LookbackFeatures}(\hat{y}_i, \ell^*))$$

The final output is chosen by selecting the candidate with the highest predicted score:

$$\hat{y} = \arg \max_i s_i$$

3.3 Baseline

As a baseline, we use the original lookback Lens decoding framework [4], which scores candidates solely using internal attention signals, without external retrieval grounding. Additionally, we compare against greedy decoding.

3.4 Implementation Details

We implement RELF by modifying the decoding loop in the public Lookback Lens repository¹, and extend it with a retrieval-augmented prompt construction mechanism. Our system consists of the following components:

Wikipedia Retrieval We perform real-time retrieval of factual evidence from Wikipedia using langchain’s WikipediaLoader API. Given a natural language query q , we set `load_max_docs=3` to retrieve the top 3 relevant Wikipedia articles using MediaWiki’s built-in search engine. Each document is then automatically split into smaller textual chunks using LangChain’s built-in RecursiveCharacterTextSplitter, with default parameters (chunk size ≈ 1000 characters, no overlap). This results in $\sim 10\text{--}30$ paragraph-like text chunks per query.

To better understand how the retriever positions the query relative to the retrieved context, we project the embeddings into 3D using PCA (Figure 3). This helps visualize the geometric structure of similarity decisions.

Semantic Embedding and Similarity Search To identify the most relevant evidence among the retrieved Wikipedia chunks, we encode both the input query and each chunk using the sentence-level embedding model `intfloat/e5-base-v2`, a 768-dimensional dense encoder pretrained on retrieval-specific tasks. We compute a single query embedding $\mathbf{q} \in \mathbb{R}^{768}$ and a set of chunk embeddings $\{\mathbf{d}_i\}_{i=1}^n$, and construct a FAISS index (IndexFlatL2) to perform dense vector similarity search. Although FAISS natively computes L2 distance, we apply L2-normalization to all embeddings before indexing, effectively converting the similarity metric to cosine similarity.

¹<https://github.com/voidism/Lookback-Lens>

We retrieve the top- k (default $k = 4$) most semantically similar Wikipedia chunks according to cosine similarity with the input query, and concatenate them in descending order of similarity to form the retrieval-augmented context:

$$RetrievedContext = d_{i_1} \parallel d_{i_2} \parallel \cdots \parallel d_{i_k}$$

This context is prepended to the original user query q to form the final extended prompt.

Lookback Lens Integration We preserve the original Lookback Lens guided decoding framework [4], where the model generates n candidate completions via sampling, and each candidate is evaluated based on its internal attention behavior. Specifically, we extract token-level lookback ratio vectors $\mathbf{v}_t$ by computing attention distributions over context tokens vs. newly generated tokens across all layers and attention heads. These vectors are averaged over output spans to obtain $\bar{\mathbf{v}}$, which is passed to a pretrained logistic regression classifier $\mathcal{F}$ that outputs a score indicating the likelihood of being hallucination-free.

Final Output Selection Among all generated candidates, we select the one with the highest predicted factuality score:

$$\hat{y} = \arg \max_i \mathcal{F}(\bar{\mathbf{v}}_i)$$

All RELF-specific components, including the retrieval pipeline, embedding interface, FAISS-based similarity search, prompt augmentation, and Lookback Lens-based reranking logic, were implemented from scratch and integrated with the existing decoding infrastructure.

4 Experiments

We conduct experiments to assess how RAG influences hallucination in SLMs, and whether the Lookback Lens decoding strategy can mitigate these effects. All experiments use the Natural Questions Open (NQ-Open) dataset [14], which contains real user questions and short factual answers drawn from Wikipedia. We focus on NQ-Open for its short-answer format, which makes it straightforward to identify hallucinations: a generated answer is either exactly a correct fact or clearly incorrect.

4.1 Data

Natural Questions Open is an open-domain question answering benchmark derived from the Natural Questions dataset [14]. Each question is a real Google query, and the answers are brief factual statements (often a name, date, or term). We use the standard NQ-Open split for evaluation. The short, factual nature of this dataset is ideal for our study, as it provides an unambiguous signal of hallucination – if the model’s answer does not match the ground truth, it is evidently unsupported or invented. For our retrieval-augmented experiments, we use the English Wikipedia corpus introduced above as our knowledge source.

4.2 Evaluation method

We report results using the Exact Match (EM) metric [15, 16], which is a stringent measure commonly used in QA evaluations. EM requires the model’s predicted answer to exactly match one of the ground-truth answers after normalization (i.e., ignoring minor differences in punctuation, articles, or casing). In practice, this means the model must output the correct answer verbatim to receive credit. This metric is appropriate for NQ-Open’s short answers, where any deviation usually indicates a mistake. A high EM score thus directly reflects strong factual accuracy, while a drop in EM typically signifies more frequent hallucinations or incorrect answers.

In addition to overall EM, we also analyze the nature of the model’s outputs with respect to grounding in the retrieved evidence. Specifically, we categorize each generated answer based on whether it contains hallucinated content (information not supported by any retrieved passage) and whether it is grounded in the provided context. Because NQ-Open answers are usually concise factual entities, this binary distinction is straightforward in our evaluations. We use the Lookback Lens classifier [4] as an additional tool to detect contextual hallucinations, by examining the model’s attention patterns on the context versus its own generations. However, as we will show, the presence of retrieved context complicates the effectiveness of this detector.

Query: *Who got the first nobel prize in physics?*

Retrieved Context (3 Wikipedia passages)

Doc 1 – Women in Physics: **Marie Curie** was the first woman to be nominated in 1902 and to receive the prize in 1903...

Doc 2 – Nobel Prize controversies: Since the first award in 1901, conferment of the Nobel Prize has engendered criticism and controversy.

Doc 3 – Saul Perlmutter: **Saul Perlmutter** shared the 2006 Shaw Prize in Astronomy, the 2011 Nobel Prize in Physics, and the 2015 Breakthrough Prize in Fundamental Physics with Brian P. Schmidt and Adam Riess for providing evidence that the expansion of the universe is accelerating.

LLM Output (greedy + RAG)

“Marie Curie got the first Nobel Prize ...”

Gold Answer: *Wilhelm Conrad Röntgen (1901)*

Figure 2: Failure case of RAG with a 7B LLM: retrieved passages overwhelm the model, leading to hallucination.

4.3 Experimental details

Model and Training: We conduct our tests on a **SLM, LLaMA-2-7B-Chat** model as the base SLM. We use the Hugging Face implementation (meta-llama/llama-2-7b-chat-hf) without further fine-tuning. We employ the same pretrained Lookback Lens classifier in our RAG-augmented decoding. Although retrieval supplies additional context, the 7B model’s limited parameter capacity remains a bottleneck in processing and effectively integrating the extended input.

We implement the **Lookback Lens** approach following [4]: we use a lightweight classifier that monitors the model’s attention at each decoding step. We conducted our experiments using the original Lookback Lens classifier trained on the CNN/DM [17] dataset, as provided in the official implementation. During decoding, if the classifier indicates a high probability of hallucination, the generation is guided accordingly (e.g., by resampling or adjusting the probability distribution). All decoding runs were executed on an NVIDIA GeForce RTX 4090 GPU (24GB). This classifier-guided decoding (Lookback Lens decoding) aims to steer the model towards using the context and away from unsupported content.

Compared Settings: We evaluate four generation configurations to isolate the effects of RAG and Lookback Lens:

1. **Greedy (no RAG):** Baseline greedy decoding with no retrieved passages. The model sees only the question and generates an answer by always choosing the highest-probability next token.
2. **Lookback Lens (no RAG):** Lookback Lens-guided decoding on the question alone. The classifier intervenes if the model starts to stray beyond its learned knowledge (since no external context is provided).
3. **Greedy + RAG:** Greedy decoding with retrieval-augmented input. The model receives the question plus 3 retrieved Wikipedia passages as context, and decodes greedily.
4. **Lookback Lens + RAG (RELF):** Lookback-guided decoding with retrieval. The model gets the same retrieved context, and the Lookback Lens detector guides the generation to avoid context-independent hallucinations.

These four settings allow us to examine two questions: (a) Does adding a retrieval context help or hurt the SLM’s factual accuracy (b) Can the Lookback Lens detector improve the SLM’s performance, especially in the presence of retrieval.

4.4 Results

Table 1 reports the main EM results for all settings. Without any retrieval, the SLM performs reasonably well: the **greedy baseline** achieves 0.732 EM score, and the **Lookback Lens decoding** is similar at 0.718. This small difference suggests that in the standard (no-RAG) scenario, the classifier-guided decoding neither significantly improves nor harms accuracy in this experiment—in

Method	Best span EM
Greedy	0.7318
Greedy + RAG	0.3925
Lookback	0.7182
lookback + RAG (RELF)	0.3845

Table 1: Exact match (EM) scores of different decoding and retrieval configurations. Combining RAG with either decoding strategy notably decreases performance.

Setting	Both Wrong	RAG Harmed	RAG Helped	Both Correct
Greedy	0.2056	0.4019	0.0625	0.3299
Lookback	0.2110	0.4028	0.0637	0.3225

Table 2: Proportion of examples by outcome category comparing RAG vs no-RAG predictions. “RAG Harmed” refers to cases where only no-RAG was correct, while “RAG Helped” denotes cases where only RAG was correct.

the previous paper [4], Lookback Lens was about three percentage points higher than the greedy baseline.

When we add retrieval, the **Greedy + RAG** model’s EM plummets to 0.393, roughly half the no-RAG performance.

Importantly, even Lookback Lens decoding does not rescue performance in the RAG setting. **Lookback Lens + RAG (RELF)** yields an EM score of 0.385, virtually the same (in fact slightly lower) than **Greedy + RAG**. These EM scores confirm that Lookback Lens + RAG (RELF) offers no gain over the unguided RAG decoder in terms of producing correct answers.

Table 2 breaks down, for each question, whether the RAG-based answer and the non-RAG answer were correct or hallucinated, yielding four outcome categories:

- **Both Wrong:** neither RAG nor non-RAG answer is correct.
- **RAG Harmed:** non-RAG is correct but RAG is wrong.
- **RAG Helped:** RAG is correct but non-RAG is wrong.
- **Both Correct:** both RAG and non-RAG answers are correct.

For example, under greedy decoding, 40.19% of questions fall into “RAG Harmed,” meaning retrieval caused the model to switch from a correct no-RAG answer to an incorrect one, while only 6.25% are “RAG Helped.” Lookback Lens shows a near-identical pattern—40.28% harmed versus 6.37% helped—demonstrating that the error introduced by retrieval remains unchanged even with attention-based guidance. This reinforces that retrieval more often degrades than improves performance on our SLM, and that Lookback Lens fails to fundamentally alter the distribution of RAG-induced hallucinations.

5 Analysis

This section inspects *why* adding retrieval so dramatically degraded performance and why Lookback Lens failed to mitigate the problem. We advance two complementary claims.

Claim 1: Capacity/Length Constraint We attribute this to capacity limitations: the 7B SLM cannot effectively filter and integrate the *three full Wikipedia articles* we supply, leading to confusion and an elevated hallucination rate. Thus, Lookback’s hallucination-mitigation mechanism cannot substantially reduce the additional hallucinations caused by RAG; it merely softens the overall accuracy loss.

Claim 2:Retriever Quality Even if the SLM *could* cope with long contexts, a flawed retrieval stage can still poison the input. Our retriever ranks passages solely by **L2 similarity of sentence embeddings**, with no constraint that the returned text actually *answers* the question. Figure 2 shows the consequence: for “*Who got the first Nobel Prize in physics?*” the top-*k* passages mention Marie Curie (1903) and Saul Perlmutter (2011) but never Wilhelm Conrad Röntgen (1901). Because the highest-ranked chunk explicitly says Curie received the prize in 1903, the language model confidently hallucinates that she was the first laureate.

Figure 3 provides geometric insight. It plots a PCA projection of the query embedding (red sphere) and its three nearest neighbours in the 768-dimensional space. The query lies closer to Doc 1 than to any snippet containing “Röntgen,” so a purely distance-based retrieval policy inevitably omits the correct passage. In short, *semantic proximity does not guarantee answer relevance*: dense retrieval without an answer-aware objective frequently surfaces thematically related but answer-free text, forcing the decoder to choose among distractors and thereby amplifying hallucinations.

Collectively, capacity constraints and retrieval-quality deficits impose compounded performance bottlenecks on SLMs. We therefore conclude that *improved retrieval quality is as crucial as capacity-aware decoding* when applying RAG to SLMs.

6 Conclusion

Building on prior work showing that RAG reduces context hallucination in language models and that Lookback Lens already curbs such errors in SLMs, we ask a compositional question: does applying Lookback Lens to an SLM after its prompt has been expanded with retrieval yield an additive accuracy gain? To test this, we compare four decoding setups—Greedy, Greedy + RAG, Lookback Lens-Guided, and Lookback Lens-Guided + RAG—and assess whether the separate strengths of RAG (external grounding) and Lookback Lens (hallucination detection) reinforce each other when combined.

The results contradicted our initial hypothesis. EM followed the order Greedy: 0.732 > Lookback Lens Guided: 0.718 (−1.4 pp) » Greedy + RAG: 0.392 > Lookback Lens Guided + RAG: 0.385 (−0.7 pp). In other words, adding RAG reduced the SLM’s factual accuracy by roughly 34–35 percentage points, and Lookback Lens Guided decoding altered performance by only a few percentage points regardless of RAG. Lookback Lens therefore failed to detect or mitigate hallucinations introduced by RAG.

These findings point to two complementary factors which explain why adding RAG sharply degraded performance and why Lookback Lens could not recover the hallucination. First, model capacity and length constraint. A 7B-parameter SLM cannot reliably filter or integrate the three full Wikipedia articles inserted by RAG. The oversized prompt confuses the model, raising the hallucination rate and leaving little room for any decoding heuristic (Lookback Lens) to restore accuracy. Second, retriever quality deficit. Our retriever ranks passages solely by embedding similarity, so top-*k* results often omit the answer altogether and instead return thematically related but answer-free text. Because Lookback Ratio measures only where the model attends—not whether the attended tokens are correct—the metric remains in a normal range even when the prompt itself contains false or irrelevant evidence. The SLM then confidently repeats those errors.

A primary limitation of this study is that all experiments were conducted on a single SLM architecture and parameter scale. The findings therefore may not generalize to larger or differently trained small-capacity models, nor to alternative retrieval pipelines or hyper-parameter settings.

Future work should address 1) adaptive RAG filtering or compression tailored to SLM capacity, 2) a retrieval-aware Lookback Lens that combines retrieval confidence and source alignment with the lookback ratio, and 3) large-scale analyses that systematically vary model size, lookback hyper-parameters, and retrieval quality to categorize failure modes. Such research will clarify the limits of external evidence in SLM settings and refine strategies for hallucination mitigation.

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Appendix: Team contributions

A.1 Taeyun, Roh (Leader)

- Code: Retrieval implementation, PCA Visualization
- Report: Abstract, Introduction, Related Work, Approach, Appendix

A.2 Junhwan, Heo (Member)

- Code: Evaluation (overall), Rag Implementation

A.3 Dongyun, Nam (Member)

- Code: Rag Implementation
- Report: Experiments, Result, Analysis

A.4 Minuk, Park (Member)

- Code: Rag implementation
- Report: Analysis, Conclusion

B Background: Lookback Lens for Hallucination Detection

To detect contextual hallucinations in language models, **Lookback Lens** [4] introduces a decoding-time signal based on attention distributions. Rather than relying on output tokens directly, Lookback Lens leverages the model’s internal attention maps to estimate whether a generated span is well-grounded in the input context.

B.1 Lookback Ratio Computation

Let a transformer model have L layers and H attention heads per layer. For each decoding step t , given a prompt consisting of input tokens $X = \{x_1, x_2, \dots, x_N\}$ and generated tokens $Y = \{y_1, \dots, y_{t-1}\}$, Lookback lens compute attention-based scores for how much the model attends to context tokens versus newly generated tokens.

For each head h in layer l , the average attention to context tokens and new tokens at step t is computed as:

$$A_t^{l,h}(\text{context}) = \frac{1}{N} \sum_{i=1}^N \alpha_{h,i}^l, \quad A_t^{l,h}(\text{new}) = \frac{1}{t-1} \sum_{j=N+1}^{N+t-1} \alpha_{h,j}^l$$

Here, $\alpha_{h,j}^l$ denotes the softmax-normalized attention weight from the current token to token j in head h of layer l .

Using these, the **Lookback Ratio** at time t for head h in layer l is defined as:

$$LR_t^{l,h} = \frac{A_t^{l,h}(\text{context})}{A_t^{l,h}(\text{context}) + A_t^{l,h}(\text{new})}$$

This value reflects how much attention is paid to the context vs. to the model’s own previous generations. Higher values indicate greater contextual grounding.

B.2 Hallucination Classification

To use the lookback ratios as input features for hallucination detection, we concatenate them across all layers and heads into a feature vector for time step t :

$$\mathbf{v}_t = [LR_t^{1,1}, LR_t^{1,2}, \dots, LR_t^{L,H}]$$

Given a span of T consecutive tokens $\{y_t, y_{t+1}, \dots, y_{t+T-1}\}$, we average the lookback feature vectors over the span to obtain:

$$\bar{\mathbf{v}} = \frac{1}{T} \sum_{i=t}^{t+T-1} \mathbf{v}_i$$

A logistic regression classifier $\mathcal{F}$ is then used to predict whether the span is factual ($y = 1$) or hallucinated ($y = 0$):

$$P(y = 1 \mid \bar{\mathbf{v}}) = \mathcal{F}(\bar{\mathbf{v}}) = \sigma(\mathbf{w}^\top \bar{\mathbf{v}} + b)$$

Here, σ denotes the sigmoid function, $\mathbf{w}$ is the weight vector, and b is the bias term of the classifier. This classifier is trained using labeled spans (either via predefined annotations or heuristics), and used at decoding time to guide selection of more grounded output candidates.

C Additional Retrieval Visualisation

Why a PCA plot? To verify that FAISS is selecting passages that are *indeed* closest to the query in embedding space, we project the query vector and its k neighbours onto the first three principal

3-D Visualization of Query + Top-k Docs

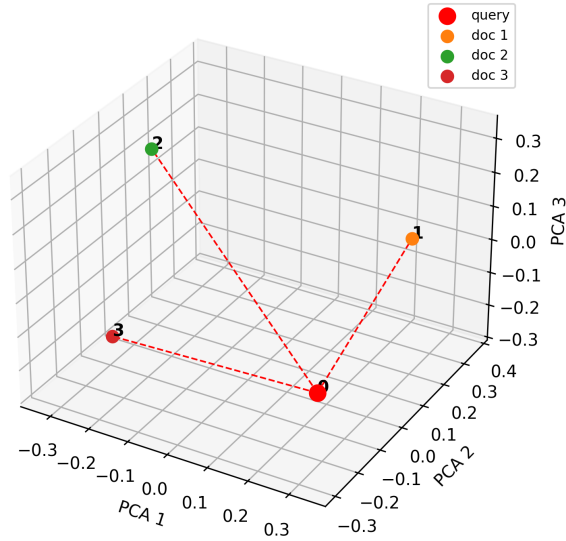


Figure 3: PCA projection of the query and top-3 retrieved Wikipedia passages in the embedding space. Each point represents a 768-dimensional sentence embedding projected into 3D using PCA. The red sphere denotes the query embedding, and dashed red lines connect it to each of the top-3 documents returned by FAISS. Shorter connections indicate higher semantic similarity in the original embedding space. This visualization helps interpret how the retriever selects passages based on embedding proximity.

components of the joint embedding matrix. Although the axes (**PCA 1–3**) have no direct linguistic interpretation, they explain the largest portions of variance so relative distances and angles remain visually meaningful. Figure 3 in the appendix shows that *doc 1* lies nearest to the query, while *doc 2* and *doc 3* occupy progressively more distant positions—mirroring their FAISS ranks.

Appendix placement. Because this plot serves only as a qualitative sanity check and does not affect the core claims, we relegate it to **Appendix C**.